

Ultra-low noise all polarization-maintaining Er fiber-based optical frequency combs facilitated with a graphene modulator

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Abstract: High bandwidth carrier phase and repetition rate control are critical for the construction of low phase noise optical frequency combs. Here we demonstrate the use of a graphene modulator for the former and a bulk electro-optic modulator for the latter enabling record low phase noise operation of an Er fiber frequency comb. For applications that do not require carrier phase control, we show that the form factor of a fiber comb can be reduced by adapting a graphene modulator for rapid repetition rate control. Moreover, the whole system demonstration is performed with all-polarization maintaining Er fiber frequency combs, highly suitable for applications in the field.

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1. Introduction

Optical frequency combs, which consist of hundreds of thousands of longitudinal modes of order n , characterized by a frequency spectrum $f_n = f_{\text{ceo}} + n f_{\text{rep}}$ with a carrier envelope offset frequency (f_{ceo}) and a frequency spacing set by the laser repetition frequency (f_{rep}), have been applied to many technology areas comprising for examples; optical atomic clocks, ultra-precise metrology, microwave generation, communication, spectroscopy, imaging and astronomical calibration [1, 2]. Indeed, optical frequency combs have evolved into an indispensable tool for many research laboratories. More recently, driven by emerging real world applications, the demand for fieldable optical frequency combs, has been growing. To fully exploit the capabilities of frequency combs, robust, portable, low phase noise optical frequency references are required, which have also greatly improved during the last ten years [3, 4]. Fieldable optical frequency combs with low phase noise are bound to further revolutionize measurement technology and to make it possible to exploit these advances outside of research laboratories. Oftentimes comb applications not only require low phase noise, but also low amplitude noise in order to be competitive with alternative technologies. Radar and navigation applications further rely on precision frequency division from the optical to the microwave domain [5–7].

Optical frequency combs based on Ti:sapphire solid-state lasers have so far generated the lowest f_{ceo} phase noise at about 0.045 rad [8]. However, Ti:sapphire optical frequency combs are difficult to operate under the influence of demanding environments because of their limited potential for integration. Diode-pumped solid state lasers have also achieved impressive phase noise performance [9, 10] but may have limited application potential for some demanding applications such as communications or space missions, because they are relatively bulky and very sensitive to environment. Fiber-based optical frequency combs based on nonlinear polarization rotation (NPR) [11] or saturable absorber (SA)-assisted NPR mode-locking mechanisms have proven to be viable alternatives and have also been operable at low values of phase noise, i.e. the lowest f_{ceo} phase noise values achieved for fiber combs to date are 0.15 rad for Tm frequency combs [12], 0.32 rad for Yb frequency combs [13], which is estimated from the coherent carrier power of the phase locked f_{ceo} as shown in [13], and about 0.4 rad for Er frequency combs [14], 0.4 rad for Er frequency combs [14], which is estimated from the PSD of the phase locked f_{ceo} as shown in Fig. 3(b) of [14]. However, all these results have been obtained in non-polarization maintaining fiber cavities, which are subject to environmentally induced fluctuations of performance. Consequently, polarization-maintaining (PM) fiber-based optical frequency combs have been long sought for comb applications in the real world [15]. Although there are a few reports about all-PM frequency combs, so far the f_{ceo} phase noise has been > 1 rad, limiting their application potential in high precision applications [16, 17].

Two fast modulators are crucial for optical frequency combs with ultra-low phase noise because of their two degrees of freedom; f_{ceo} and f_{rep} . Typically, pump current modulation is used for stabilization of f_{ceo} and an electro-optic modulator (EOM) is used to stabilize f_{rep} [14, 18–20]. However, the feedback bandwidth for pump current modulation is limited by the response time of the gain medium [21]. For example, the feedback bandwidth for pump current modulation for Er fiber lasers is around 100 kHz, which hampers ultra-low phase noise operation. Alternatively, as shown in [8], an extra-cavity acousto-optic modulator (AOM) can be used for f_{ceo} control, though the limited diffraction efficiency and induced spatial dispersion can be an issue, causing the modulation-frequency dependent loss. This would generate additional intensity noise and modulation-frequency dependent changes of the signal-to-noise ratio for the f_{ceo} . More importantly, the response time of an extra-cavity AOM is limited by the propagation delay of the acoustic waves in the crystal, resulting in around 250 kHz feedback bandwidth [22].

For many applications, such as THz or difference frequency generation (DFG), as well as Raman spectroscopy, only f_{rep} control is required, since the generated frequencies are independent of f_{ceo} . For these purposes, a fast modulator, which has low loss and is easy to install in oscillators, is sought rather than a relatively unwieldy waveguide or bulk EOM. Note that an extra-cavity AOM cannot be used for repetition rate control.

In this work, we demonstrate that a graphene based EOM can work as a high bandwidth modulator not only for f_{ceo} , but also for f_{rep} in an all PM Er fiber optical frequency comb, including an oscillator, an amplifier, and highly nonlinear fiber (HNLF). This may initially seem counterintuitive, since a graphene modulator induces frequency shifts of f_{ceo} and nf_{rep} via rapid cavity loss modulation [12, 23] that cancel near the optical carrier frequency, resulting in no overall shift of these comb teeth. However, we can stabilize f_{rep} by locking to a beat frequency (f_{beat}) made by subtracting the measured f_{ceo} from an optical beat (f_{beat}) between one of the comb teeth and a CW reference laser by a double balanced mixer. This locks nf_{rep} to the CW reference, while allowing f_{ceo} to change freely, as discussed in section 4.2. This means that the fast response of a graphene modulator (GM) can be applied to f_{rep} , bypassing the limit of the gain dynamics of the laser medium. This enables single servo loop control of frequency combs at higher modulation bandwidths than methods relying on pump current modulation [24] or even bulk EOMs, particularly for comb systems operating at high repetition rates. As pointed out in section 3, GMs are compatible with any laser operating from the visible to the mid IR; therefore this scheme can also be adapted to ultra-compact, high repetition rate, solid-state combs.

Moreover, for applications that require stabilization of both f_{ceo} and f_{rep} , a GM and an EOM can be used to simultaneously stabilize f_{ceo} and f_{rep} , respectively, with little cross talk and very high modulation bandwidths. Because of the availability of two fast modulators, record low integrated phase noise of both f_{ceo} (0.25 rad from 10 Hz to 3 MHz) and f_{beat} (0.11 rad from 10 Hz to 3 MHz) are obtained.

2. Laser configuration

For an all PM configuration, a Fig. 8 laser based on a nonlinear loop mirror [16] for passive amplitude shaping or a laser mode locked with a semiconductor SA [17] are suitable. Since operation at high repetition frequencies (e.g. > 200 MHz) and incorporation of a GM are difficult for a Fig. 8 laser, we opted for a linear cavity with SA. The schematic of the oscillator is shown in the dotted green line in Fig. 1(a). The cavity is comprised of a length of non-doped PM fiber and Er-doped PM fiber with a net overall cavity dispersion of about -12000 fs^2 , ignoring the dispersion of the SA and GM. The oscillator was pumped from 100 mW with a laser diode operating at 976 nm. To minimize phase noise by reducing the cavity loss, a 5% fiber output coupler was employed. The polarization beam splitter prevents oscillation along the fast axis of the PM fiber. The use of a SA with sub ps carrier lifetime was critical to enable stable laser oscillation at small dispersion values [15]. Extended lifetime testing of the SA revealed no noticeable SA degradation over months of operation.

The output power from the oscillator is about 1 mW. The repetition frequency is about 80 MHz. As shown in [25], a repetition rate of around 80 MHz is a good compromise in terms of laser stability and achievable phase noise in microwave applications. The maximum spectral bandwidth obtained from the oscillator is about 20 nm [Fig. 1(b)], which corresponds to about 130 fs of Fourier transform-limited pulse duration, assuming a sech^2 pulse shape.

An intracavity EOM based on lithium niobate was used for high bandwidth repetition frequency control and a GM for carrier phase control. A piezo-electric transducer (PZT) was further implemented for slow f_{rep} control on the side of the fiber. The graphene modulator was designed with a modulation range of 2.5% for an insertion loss of 10% and is further described in the next section. The output from the oscillator is divided into two parts. The first part is used as a monitor for phase locking of the frequency f_{beat} between the comb and a single-longitudinal mode CW laser with specified 3 kHz linewidth. The second part is amplified to 100 mW by a PM positive dispersion Er fiber amplifier. The Er amplifier is pumped from both directions with forward and backward pump powers of 200 mW and 600 mW, respectively.

The spectrum is broadened in the amplifier; the combination of two lengths of standard negative dispersion communications fiber before the Er-doped fiber (EDF) and after the EDF is optimized to obtain the shortest pulse duration before the HNLF. A pulse duration of below 40 fs, measured by an autocorrelator, is obtained. An octave spanning spectrum is obtained after propagation through 50 cm of HNLF. The f_{ceo} was detected with a standard f-2f interferometer; two filters with 2 and 10 nm bandwidth were inserted to filter out the second harmonic signal at 1100 nm. A short piece of undoped PM fiber after the HNLF is used to adjust the group delay between 1100 nm and 2200 nm downstream from the frequency doubling crystal (periodically poled LiNbO₃). The carrier envelope offset frequency was detected with more than 35 dB SNR at 100 kHz resolution bandwidth [Fig. 1(c)], though we have also observed >40 dB SNR for slightly longer oscillator pulses. The linewidth of the free-running f_{ceo} is about 40 kHz, which is the narrowest of any Er mode-locked laser with a SA [26, 27] to date and comparable to Er fiber lasers based on nonlinear polarization rotation [14, 19].

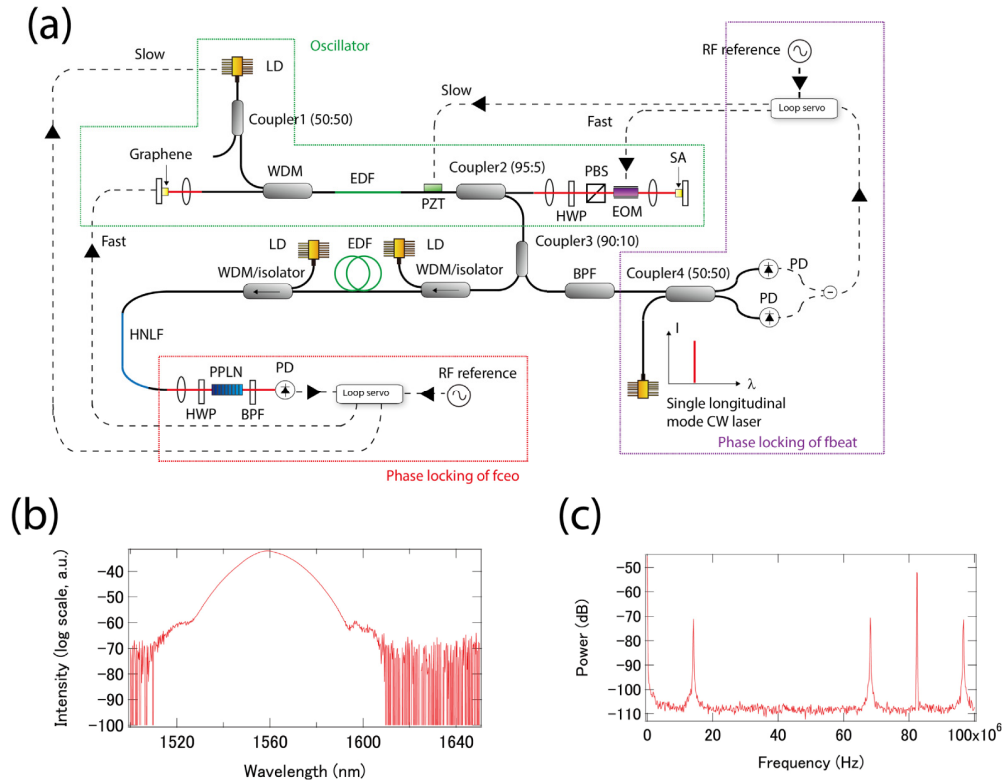


Fig. 1. (a) Schematic of laser system. LD, laser diode; WDM, wavelength division multiplexer; EDF, erbium-doped fiber; PZT, piezoelectric transducer; HWP, half waveplate; PBS, polarization beam splitter; EOM, electro-optic modulator; SA, saturable absorber; BPF, bandpass filter; PD, photo detector; HNLF, highly nonlinear fiber; PPLN, periodically-poled LiNbO₃. (b) Optical spectrum from the oscillator. (c) RF spectrum of free-running f_{ceo} with 100 kHz RBW.

3. Graphene modulator

The graphene modulator [12, 23] used in this experiment, as shown in Fig. 2, was fabricated using graphene grown on copper foil by chemical vapor deposition. The growth started with evacuating a growth chamber made from a quartz tube to $\sim 10^{-6}$ Torr while the chamber was heated to 1000°C. Then hydrogen and methane were flowed into the tube while the temperature was maintained at 1000°C for 5 minutes, after which the chamber was cooled down passively at room temperature with only hydrogen flowing. Next, the copper foil with graphene was spin-coated with polymethyl methacrylate (PMMA) on one side before the copper was etched by ferric chloride solution. The etchant solution was repeatedly diluted by deionized water, and then with the PMMA/graphene layer floating in the water, the substrate was brought into contact with it, completing the transfer process. The substrate consists of a silicon wafer with an oxidized layer, a 100 nm gold layer as the bottom contact (applied via thermal evaporation), and ~ 120 nm of tantalum pentoxide as a high-k gate dielectric layer (applied via reactive sputtering with a pure tantalum target). After the graphene is transferred onto the substrate, PMMA was removed by acetone and a metallic top contact layer was deposited by thermal evaporation (20 nm of Ti and 100 nm of Au), which was patterned by a liftoff process.

Although the absorption of monolayer graphene is only 2.3%, one can tailor the total modulation depth and insertion loss of the modulator by changing the thickness of the tantalum pentoxide gate dielectric. Since the thickness of graphene is only one atomic layer,

its modulation effect on the change of optical path length is to the first order negligible, whereas the absorption change can be higher than 50% of the total absorption. For this particular fiber oscillator, we chose the thickness of tantalum pentoxide to be 120 nm, yielding ~10% total insertion loss (5% from the bottom metallic contact; 5% from graphene) and ~2.5% modulation depth. An increase in modulation depth generally increases the f_{ceo} tuning range with minimal effect on modulation bandwidth. In the present system the diameter of the aperture for the GM was chosen as 700 μm , which, with the contact resistance of metal on graphene, and coupled with the capacitance due to the large aperture, gave an RC-limited 3dB modulation bandwidth of around 2.5 MHz. We fabricated several such modulators and have not observed any degradation of performance even after 1 year of operation under ambient conditions.

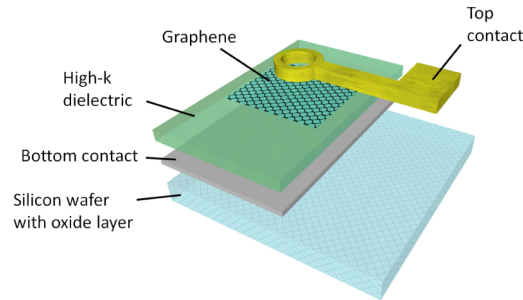


Fig. 2. Sliced structure of the graphene modulator. The bottom contact, a thin layer of metal with high reflectance (silver or gold for near IR light), serves both as an electrical contact and a highly reflective mirror. The thickness of the high-k dielectric determines the intensity of light experienced by graphene, thus providing design freedom in the insertion loss and modulation depth of the device, despite the universal broadband 2.3% absorption of graphene.

We estimate that GMs can be used for intra-cavity loss modulation in the whole near IR spectral range from 0.7 to 2.5 μm . At the short wavelength extreme, GMs can be limited in their modulation depth due to the large mismatch between their Fermi level (typically 0.2 eV below the Dirac point) and the energy of the states that absorb light (at half the photon energy below the Dirac point, which is ~0.9 eV for 0.7 μm light). At the long wavelength extreme, absorption saturation at high illuminating intensity may potentially be a limitation, but in the present system we operated with an intensity on the GM of around 3.5 MW/cm^2 , which is only around 1 - 2% of its saturation intensity of $I_{\text{sat}} = 90\text{-}400 \text{ MW}/\text{cm}^2$ at 1550 nm, as has been measured by several research groups [28–30] and as predicted by rate equation calculation [31]. According to the same theory [31] and a summary of measured saturation intensity at different wavelengths [32], the saturation intensity decreases with increasing wavelength, which may make the saturable absorption observable in GMs for wavelengths longer than 2.5 μm . However, this can in principle be overcome by increasing the beam size on the GM, or by choosing an optimum thickness of the high-k dielectric such that the field amplitude in the graphene layer is low, with the compromise of lower modulation depth.

4. Phase locking

4.1 f_{ceo} and f_{beat}

The carrier envelope offset frequency was detected as shown in Fig. 1(c). For stabilizing f_{rep} , an optical beat between one of the comb teeth and a single longitudinal mode CW laser is obtained by interfering the two lasers through a 50:50 coupler. A 40 dB SNR at 100 kHz resolution bandwidth is obtained.

Ideally, cross talk between the two modulators should be small to tightly stabilize both f_{ceo} and f_{beat} . A GM modulates f_{ceo} and $n f_{\text{rep}}$ by almost the same amount, but with opposite signs, i.e. its fixed point [33] is very close to the optical carrier frequency, shown schematically in Fig. 3, which is experimentally confirmed by adding the sinusoidal modulation to the GM

while observing signals; f_{ceo} , f_{beat} , and nf_{rep} . Here, n is an integer and, in our case, about 2.4×10^6 , with nf_{rep} being the optical carrier frequency (~ 190 THz). On the other hand, the modulation of f_{ceo} by an EOM is 100 times smaller than the modulation of nf_{rep} [34], which indicates that the fixed point of the EOM is close to f_{ceo} . This means that the combination of a GM and an EOM for highly desirable orthogonal control and phase locking of both f_{ceo} and f_{beat} at high modulation bandwidths is possible.

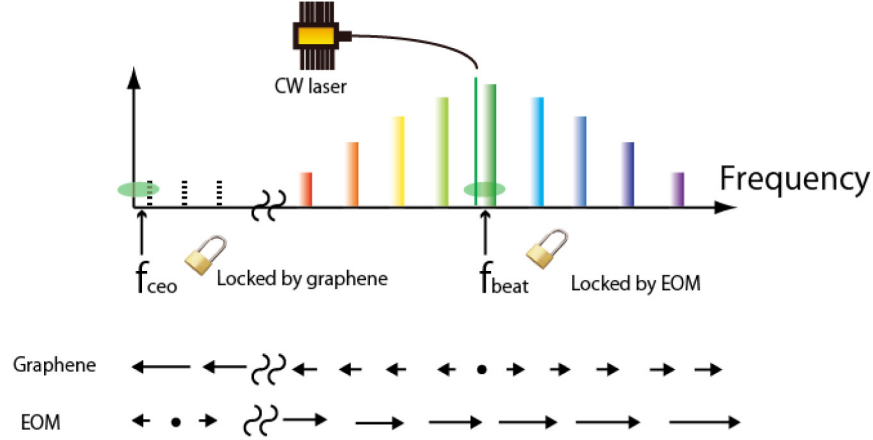


Fig. 3. Schematic for locking of f_{ceo} and f_{beat} . f_{ceo} is observed by the self-referencing scheme, and locked by a GM. f_{beat} is the beat between one of the comb teeth and a single longitudinal mode CW laser and locked by an EOM. The fixed point of a GM is near the optical carrier frequency. The fixed point of an EOM is near f_{ceo} .

Since the modulation depth of a GM is not sufficient to capture the appropriate beat for f_{ceo} control, we also employ pump current modulation as a slow modulator for f_{ceo} . To stabilize f_{beat} , it is possible to use only the EOM, at least in principle, but the halfwave voltage for a bulk-EOM is very high (typically a few 100 V), which limits the feedback bandwidth. Thus, a PZT, with a large modulation depth, is used as a slow cavity length modulator, and the EOM is used as a fast modulator.

To show the advantage of a GM compared with pump current modulation for f_{ceo} control, the amplitude and phase response for optical power modulation by the GM and pump current modulation are measured [Figs. 4(a) and 4(b)]. The response of the optical power should be very similar to the response of f_{ceo} as shown in [35], because any optical power change by cavity loss modulation influences nonlinear effects in the cavity, such as SPM or self-steepening, resulting in a change of f_{ceo} . Amplitude modulation at high frequencies is mostly unaffected by pump current modulation because of the lifetime of Er , which works as a low pass filter. The amplitude response of a GM is significantly larger at high frequencies. Furthermore, phase rotation for pump current modulation is larger than for a GM, which makes it difficult to design a loop filter for feedback.

Figure 3 (c) shows the RF spectra of the locked f_{ceo} . The servo bump induced by pump current modulation is at 150 kHz. On the other hand, the servo bump due to loss modulation via the GM exceeds 1 MHz, which improves the coherent peak from 30 dB to 45 dB at 3 kHz RBW. The integrated in-loop phase noise of the locked f_{ceo} by loss modulation is about 0.25 rad from 10 Hz to 3 MHz [Fig. 3(e)]. For f_{beat} , the RF spectrum and power spectral density (PSD) are shown in Fig. 3(d) and 3(f). Some spikes caused by mechanical resonances of the EOM appear at 300 kHz and 1 MHz. The obtained integrated in-loop phase noise from 10 Hz to 3 MHz is as small as 0.11 rad. The obtained phase noise of f_{beat} is smaller than that of f_{ceo} , because the intrinsic phase noise of f_{beat} is smaller than that of f_{ceo} . Note that we did not observe any difference of the f_{ceo} RF spectrum with and without phase locking of f_{beat} , and vice versa, and the shown RF spectra and PSDs are taken when both f_{ceo} and f_{beat} are phase locked simultaneously.

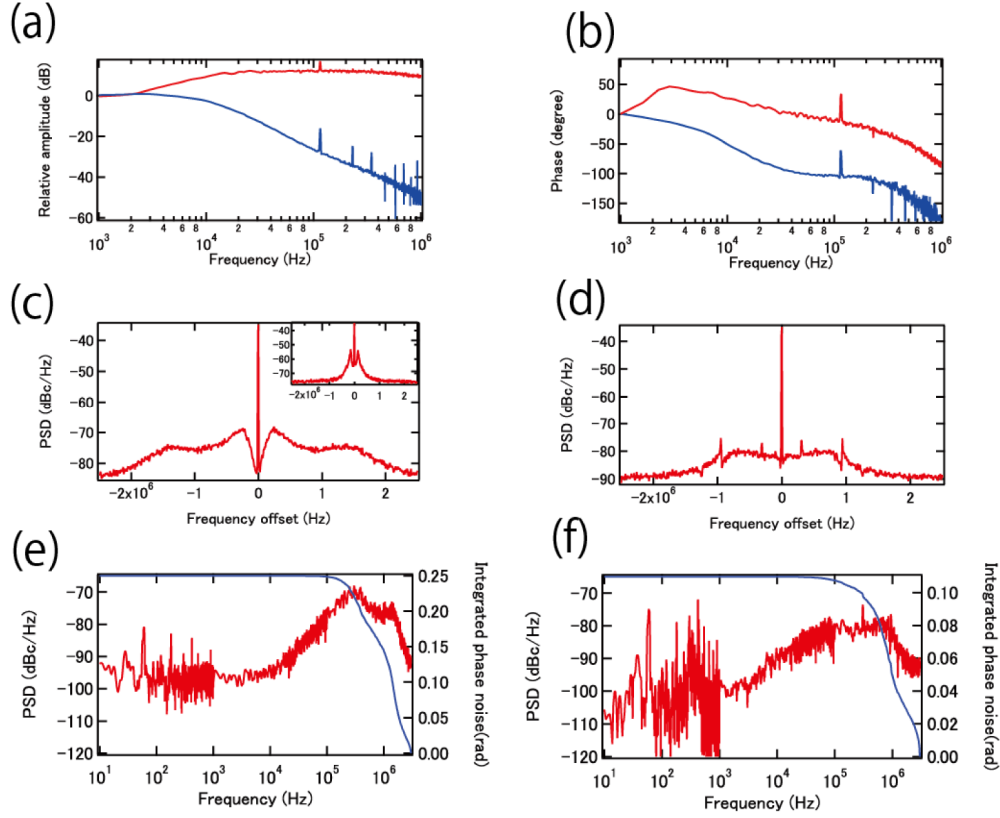


Fig. 4. Transfer function of graphene modulator (GM) (red) and pump current modulation (blue) for optical power, in (a) amplitude and (b) phase. (c) RF spectrum of locked f_{cco} by GM with 3 kHz RBW. The inset is the RF spectrum of locked f_{cco} by pump current modulation with 3 kHz RBW. (d) RF spectrum of locked f_{beat} by an EOM with 3 kHz RBW. (e) PSD of locked f_{cco} . (f) PSD of locked f_{beat} . (b, c, d, and e) are in-loop measurements.

4.2 nf_{rep}

Since many application require only precise stabilization of f_{rep} , a compact alternative to conventional EOMs for fast repetition frequency modulation is highly sought. Here, we show that a GM can also work to stabilize f_{rep} . To access f_{rep} in the optical domain, the f_{cco} is electronically subtracted from f_{beat} [7, 24];

$$f_{\text{beat}}' \equiv nf_{\text{rep}} - f_{\text{cw}} = f_{\text{beat}} - f_{\text{cco}} = (nf_{\text{rep}} + f_{\text{cco}} - f_{\text{cw}}) - f_{\text{cco}} \quad (1)$$

This scheme is particularly useful for single servo loop control of frequency combs. f_{beat}' does not depend on f_{cco} , and nf_{rep} is phase-locked to f_{cw} by stabilizing f_{beat}' . Experimentally, this is realized by mixing f_{beat} and f_{cco} in the RF domain by a double balanced mixer. Although the actual locking results depend on the setting of the loop filter and the applied voltage to the EOM, f_{beat}' locking by a GM is better than by an EOM with respect to the obtainable coherent peak signal to noise ratio. The coherent peaks of locked f_{beat}' by graphene and EOM are 40 dB and 35 dB at 3 kHz RBW, respectively, as shown in Fig. 5(a) and 5(b). From Fig. 5(a), the carrier power of locked f_{beat}' with the GM is estimated to be 90%, resulting in 0.32 rad integrated in-loop phase noise from DC to 2 MHz. Since f_{cco} is completely free-running, and a fixed RF filter is used to extract f_{cco} , phase locking could not be preserved long enough to measure the PSD of the phase locked signal. This will be readily solved by loosely locking f_{cco} . Since a 1 Hz of feedback bandwidth is large enough, a software-based loop filter can be used to keep f_{cco} within an RF filter bandwidth.

We trace back the slightly worse performance of a GM when applied to f_{rep} rather than f_{ceo} locking to the difference between the free-running linewidths for f_{ceo} and f_{beat} , although, ideally, the free-running linewidths of f_{ceo} and f_{beat} should be almost the same because the fixed point of the oscillator is around the laser frequency [33]. Also since the modulation bandwidth of loss modulation depends linearly on f_{rep} [12], we expect even better performance of a GM for f_{rep} locking for a 200 or 400 MHz mode-locked laser. Note that a GM can be incorporated into a cavity with a much smaller form factor compared to an EOM. Indeed, we have completely packaged a GM and its electronic driver as an end mirror of the above laser, in an 8 mm long button with a diameter of 15 mm.

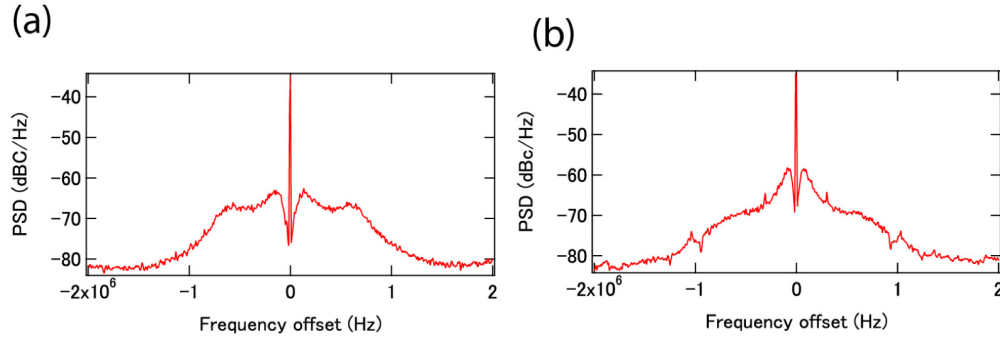


Fig. 5. (a) In-loop RF spectrum of locked f_{beat} by GM with 3 kHz RBW. (b) In-loop RF spectrum of locked f_{beat} by EOM with 3 kHz RBW.

4. Conclusion

In conclusion, we demonstrated phase locking of both carrier phase and repetition rate in a mode locked laser for the first time. The implementation of cavity loss modulation compared to pump modulation increases the carrier phase response bandwidth by around an order of magnitude; the use of a graphene modulator in conjunction with an EOM allows for high bandwidth control of both carrier phase and repetition frequency. The adaptation of a graphene modulator to the important telecom wavelength range of 1.55 μm enables the construction of compact Er fiber lasers with record low phase noise for both carrier phase and repetition frequency locking. Moreover, for applications where carrier phase control is not required, a graphene modulator can be used for repetition frequency control as an alternative to conventional electro-optic modulators while also allowing for a smaller form factor and lower phase noise. Finally, the implementation of a dispersion compensated all polarization maintaining fiber oscillator ensures a robust system design compatible with even the most challenging environments. We believe that ultra low phase noise frequency combs as demonstrated here will greatly expand the application potential of frequency comb technology.

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